

# Does Expert Sparsity Concentrate Social Bias?

## A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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### Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction  $k/N \in [0.016, 0.25]$ , and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant* at 0.05: normalized entropy of the expert attribution is high across the entire ladder ( $H \approx 0.79$ – $0.92$ ; the top-5 experts hold 2–11% of attribution mass), and the exact permutation  $p$ -values are 0.106 (all 6 rungs,  $\rho_H = +0.754$ ,  $r^2 = 0.57$ ), 0.100 (5 rungs,  $\rho_H = +0.872$ ,  $r^2 = 0.76$ ), 0.167 (4 rungs,  $\rho_H = +0.949$ ,  $r^2 = 0.90$ ), and 0.333 (3 rungs)—the exact tie-corrected attainable minimum is 0.0056 at  $n = 6$  and 0.0333 at  $n = 5$  (both well below 0.05), so the 6- and 5-rung results are genuinely underpowered rather than floor-bound; a Monte Carlo power simulation at the observed  $\rho_H$  indicates  $n \approx 8$ – $13$  ladder rungs would be needed for 80% power (Section 6.4). Dense baselines sit clearly lower ( $H \in [0.66, 0.76]$ , mean 0.70), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Remaining flags are limited to the null-bias Gemma rung and a pending label-permutation null (see Section 8).

### CCS Concepts

• **Computing methodologies**  $\rightarrow$  *Artificial intelligence*.

### Keywords

mixture of experts, bias, Shapley values, interpretability

## 1 Introduction

MoE models route each token through a subset of “expert” subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller  $k/N$ ) implies stronger attribution concentration, with Gini rising from the  $\sim 0.5$ – $0.6$

range toward 0.72–0.73 for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and  $n \leq 6$  point estimates. This round-II audit fixes the measurements, reports exact permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

- (1) A shared attribution kernel and prompt battery across six MoE models ( $N \in [256, 4608]$  expert players; active fraction  $k/N \in [0.016, 0.25]$ ) and four dense FFN baselines ( $N = 32$  layer players).
- (2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at 0.05**. All four concentration metrics (normalized entropy  $H$ , Gini  $G$ , top-fractions  $t_5$  and  $t_{10\%}$ ) move monotonically with sparsity; dense models separate as a group. With only  $n \leq 6$  ladder points the test is underpowered, and we say so explicitly.
- (3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation  $p$ -values, and an explicit model-set sensitivity audit reported for every subset so that “selecting the subsample that supports the claim” is impossible.
- (4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley flights) showing no demographic specialist structure.
- (5) Data/code release (**expt-bias-1/**).

**Take-away.** Take-away: none of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

## 2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in our prior work [15]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9]) define the routers we instrument; recent surveys [10] catalogue the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [11], BBQ [12], holistic audits

[13, 14]); our contribution measures not model-level bias magnitude but the *kernel location* of bias-relevant counterfactual shift within the routing space.

### 3 Method

#### 3.1 Routing-Shapley attribution

For each model, on contrastive prompt pairs  $(x^+, x^-)$  (stereotypical vs. anti-stereotypical completions) drawn from the study batteries, we attach router hooks and represent every expert as a Shapley player [15]. The routing-Shapley value  $\phi_i$  of expert  $i$  is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (routing\_contrast kernel, src/moe\_bias\_shapley/shapley.py). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ( $N = 32$ ).

#### 3.2 Concentration metrics

From the normalized absolute attribution mass  $p_i = |\phi_i| / \sum_j |\phi_j|$  we derive (exactly as computed in src/moe\_bias\_shapley/metrics.py):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \text{ (normalized entropy),} \quad (1)$$

$$G = \frac{N + 1 - 2 \sum_{i=1}^N \text{cum}_i(p_{(i)})}{N} \text{ (Gini of } |\phi|), \quad (2)$$

$$t_5 = \sum_{i=1}^5 p_{(i)}, \quad (3)$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \quad (4)$$

where  $p_{(i)}$  are the shares in decreasing order and  $\text{cum}_i(\cdot)$  is the cumulative sum (so  $0 \leq H \leq 1$ ). Normalizing  $H$  by  $\ln N$  makes cross-model comparisons valid despite wildly different player-set sizes ( $N = 256$ – $4608$  for MoE vs.  $N = 32$  for dense). Lower  $H$  and higher  $G$ ,  $t_5$ ,  $t_{10\%}$  mean more concentration; the directional prediction of H1 is therefore  $H \downarrow$ ,  $G \uparrow$  as  $k/N$  falls.

### 4 Experiment Setup

#### 4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero  $\phi$  (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ( $\Delta H = -0.003$ ) and is used throughout.

Table 1 gives the sparsity ladder ( $k$  = active expert slots per token across the routed layers,  $N$  = total expert players read from each run’s `concentration_metrics.n_players`),

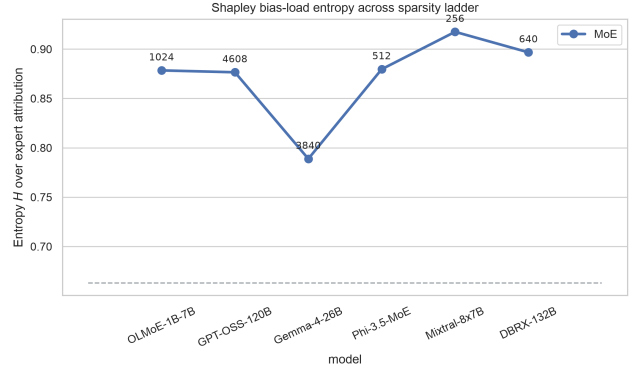


Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

along with the measured concentration metrics pulled directly from `results/*/result.json`.

### 5 Results

#### 5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with  $k/N$  that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives  $\rho_H = +0.754$  ( $p = 0.106$ , exact two-sided permutation); Gini gives  $\rho_G = -0.754$  ( $p = 0.106$ ). These are not significant at 0.05, and with  $n = 6$  ladder positions the test is simply underpowered: the tie-corrected exact-permutation minimum attainable  $p$  is 0.0056 at  $n = 6$  (well below 0.05), so this reflects a genuine power shortfall, not a discreteness floor—a Monte Carlo simulation puts 80% power at  $n \approx 12$ – $13$  rungs for this effect size (Section 6.4). Normalized entropy spans only 0.789–0.917 across the six rungs ( $H \approx 0.88$ – $0.92$  for the four un-flagged rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in  $k/N$ . By contrast, the smallest dense-to-MoE entropy gap is 0.030 (vs. the null-bias Gemma rung, the lowest MoE point; excluding that rung it is 0.117); the dense/MoE split is the robust contrast of the study.

**Take-away.** Take-away: the sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

#### 5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable “fixable” expert set.

| Model                           | $k$ | $N$  | $k/N$  | $H$   | Gini  | $t_5$ | $t_{10\%}$ | CI             |
|---------------------------------|-----|------|--------|-------|-------|-------|------------|----------------|
| OLMoE-1B-7B                     | 16  | 1024 | 0.0156 | 0.878 | 0.647 | 0.092 | 0.510      | [0.877, 0.886] |
| GPT-OSS-120B <sup>†</sup>       | 144 | 4608 | 0.0313 | 0.876 | 0.727 | 0.020 | 0.573      | [0.875, 0.886] |
| Gemma-4-26B <sup>‡</sup>        | 240 | 3840 | 0.0625 | 0.789 | 0.855 | 0.051 | 0.767      | [0.791, 0.802] |
| Phi-3.5-MoE                     | 64  | 512  | 0.1250 | 0.880 | 0.640 | 0.095 | 0.469      | [0.877, 0.892] |
| Mixtral-8x7B                    | 64  | 256  | 0.2500 | 0.917 | 0.516 | 0.107 | 0.357      | [0.913, 0.925] |
| DBRX-132B                       | 160 | 640  | 0.2500 | 0.897 | 0.599 | 0.091 | 0.436      | [0.897, 0.910] |
| Dense baselines (mean, $N=32$ ) | —   | —    | 1.0    | 0.704 | 0.682 | 0.699 | 0.595      | —              |

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03_h1_verdict.json` ladder rows).  $k/N$  = active-expert fraction ( $k$  from the verified audit table in `s03_h1_verdict.py`,  $N$  from stored `n_players`). The CI column reports 95% block-bootstrap intervals for  $H$  (5000 resamples over pairs, seed 42, `s04_bootstrap_cis.json`); intervals for Gini,  $t_5$ ,  $t_{10\%}$  are reported in Table 2. <sup>†</sup>GPT-OSS-120B v1 is a valid capture ( $n_{pairs} = 2000$ ); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ( $\Delta H = -0.003$ ). <sup>‡</sup>Gemma’s mean bias gap  $\approx 0$  (null-bias model); its concentration readings reflect routing noise, reported for transparency.

| Model                    | $H$            | Gini           | $t_5$          | $t_{10\%}$     |
|--------------------------|----------------|----------------|----------------|----------------|
| OLMoE-1B-7B              | [0.877, 0.886] | [0.630, 0.649] | [0.084, 0.096] | [0.491, 0.514] |
| GPT-OSS-120B             | [0.875, 0.886] | [0.707, 0.731] | [0.017, 0.021] | [0.540, 0.575] |
| Gemma-4-26B <sup>‡</sup> | [0.791, 0.802] | [0.841, 0.853] | [0.045, 0.051] | [0.736, 0.761] |
| Phi-3.5-MoE              | [0.877, 0.892] | [0.610, 0.643] | [0.085, 0.100] | [0.442, 0.477] |
| Mixtral-8x7B             | [0.913, 0.925] | [0.494, 0.529] | [0.098, 0.115] | [0.337, 0.368] |
| DBRX-132B                | [0.897, 0.910] | [0.569, 0.603] | [0.072, 0.090] | [0.391, 0.434] |

Table 2: 95% block-bootstrap confidence intervals for every MoE rung’s concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04_bootstrap_cis.json`). Intervals for  $H$  are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture ( $n_{pairs} = 2000$ ). <sup>‡</sup>Gemma: see Table 1.

| Model        | $H$   | Gini  | $t_5$ | $t_{10\%}$ | CI ( $H$ )     |
|--------------|-------|-------|-------|------------|----------------|
| OLMo-7B      | 0.703 | 0.719 | 0.749 | 0.562      | [0.672, 0.744] |
| Phi-3.5-Mini | 0.759 | 0.645 | 0.632 | 0.499      | [0.737, 0.781] |
| Llama-2-7B   | 0.697 | 0.654 | 0.696 | 0.648      | [0.659, 0.730] |
| Llama-3.1-8B | 0.659 | 0.708 | 0.721 | 0.669      | [0.630, 0.701] |
| Mean         | 0.704 | 0.682 | 0.699 | 0.595      | —              |

Table 3: Dense-baseline measures (v1 per-pair captures; `exp2-*-v1/per_pair_phi.npy`, one FFN layer player per transformer layer,  $N = 32$ ). The CI column reports 95% block-bootstrap intervals for  $H$  (5000 resamples over pairs, seed 42, stratified by prompt group, `s04_bootstrap_cis.json`); Gini,  $t_5$ ,  $t_{10\%}$  intervals are [0.677, 0.748], [0.693, 0.777], [0.497, 0.611] (OLMo-7B); [0.621, 0.668], [0.606, 0.653], [0.473, 0.524] (Phi-3.5-Mini); [0.624, 0.699], [0.666, 0.733], [0.605, 0.687] (Llama-2-7B); [0.669, 0.734], [0.683, 0.747], [0.622, 0.696] (Llama-3.1-8B), in that column order. Per-pair  $\phi$  payloads for dense baselines landed after the initial submission (8 GPU shard jobs, 4 models  $\times$  2 shards); point estimates shift modestly from the earlier v0 single-shot capture (e.g. OLMo-7B  $H$ : 0.719  $\rightarrow$  0.703) but the qualitative pattern is unchanged: all four dense models fall well below every MoE ladder rung’s entropy (Table 1), and CIs do not overlap the MoE band.

### 5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines ( $H$ : 0.659–0.759; mean 0.704, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it

is a pointer-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

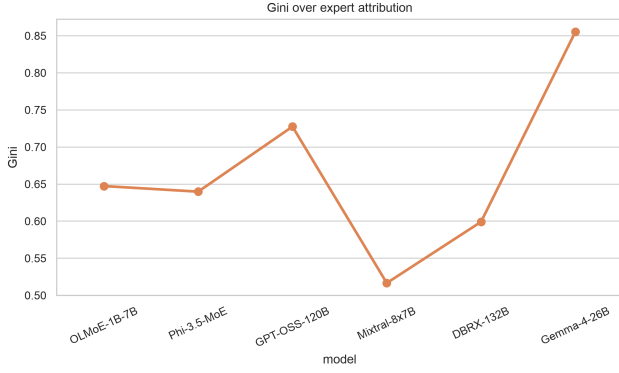


Figure 2: Gini of the expert attribution across the ladder.

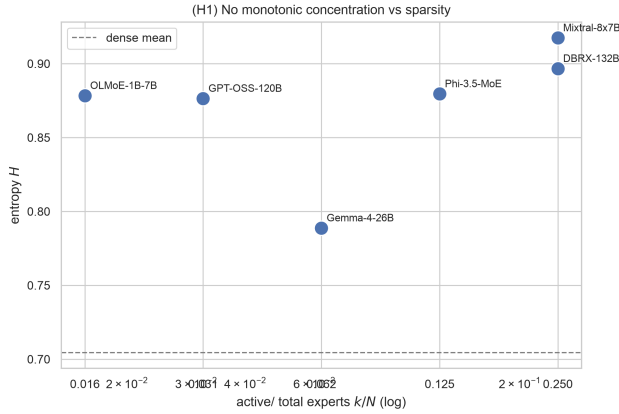


Figure 3: Entropy vs. active fraction  $k/N$  (log scale). H1 (sparser  $\Rightarrow$  more concentrated) is directionally consistent ( $\rho_H = +0.754$ , exact two-sided permutation  $p = 0.106$ ), but not statistically significant at 0.05 given  $n = 6$  ladder points.

#### 5.4 Effect sizes: geometry vs. magnitude

Two questions are conflated by significance testing alone: how large is the dense/MoE gap, and does that gap reflect a difference in bias *magnitude* or only in attribution *geometry*? Cohen’s  $d$  (pooled SD,  $n_{\text{dense}} = 4$ ,  $n_{\text{MoE}} = 6$ ) answers the first: entropy  $|d| = 3.91$  (all six MoE rungs) or 6.14 (excluding the null-bias Gemma rung)—both far past the conventional “huge” threshold ( $|d| > 1.3$ ); top-5 share  $|d| = 15.4$  (mechanical: 5 of 32 dense layers vs. 5 of 256–4608 experts); Gini  $|d| = 0.18$  (negligible—Gini does *not* separate the two groups the way entropy does). The second question has a sharp answer: the mean *absolute contrastive bias-gap magnitude* (bias\_scores.mean.bias\_gap, not a concentration metric) is 0.183 for the four dense baselines and 0.183 for the five non-null-bias MoE rungs (0.176 including Gemma), Cohen’s  $d = 0.011$  (excl. Gemma)—statistically indistinguishable.

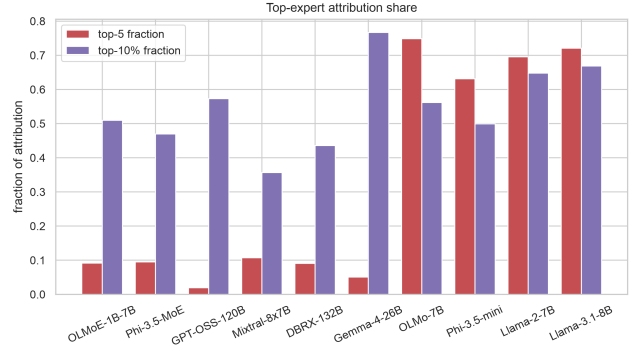


Figure 4: Top-5 and top-10% share across the ladder.

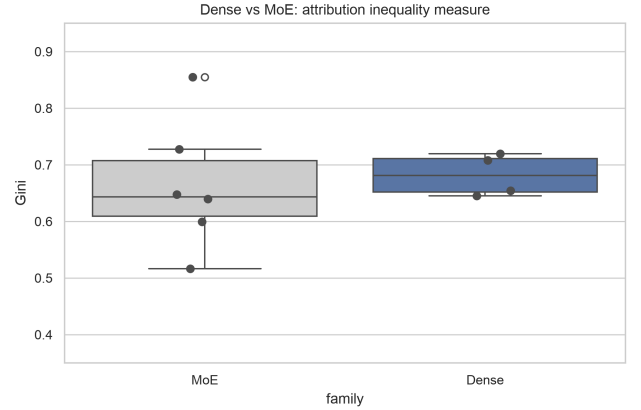


Figure 5: Dense vs. MoE Gini.

Concentration geometry and bias magnitude are therefore *orthogonal*: dense and MoE models carry the same-sized contrastive bias, but express it through entirely different attribution shapes. This is the quantitative version of the routing-structure caveat revisited in Section 7: every concentration or JS metric in this paper characterizes *where* bias-relevant shift localizes, never *how much* bias exists. We also compute a per-model *localizability ratio*  $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$  (Appendix A):  $LR \in [0.77, 0.89]$  for every MoE rung, including on matched families (OLMoE vs. OLMo-7B,  $LR = 0.800$ ; Phi-3.5-MoE vs. Phi-3.5-Mini,  $LR = 0.863$ ), ruling out unmatched architecture as an explanation for the dense/MoE split.

#### 5.5 Expert-pair synergy (Exp. 3)

The routing-Shapley decomposition reports only *marginal* attributions  $\phi_i$ ; if bias is driven by expert *pairs* acting jointly rather than by individuals, marginal attribution understates the true mechanism. We compute pairwise Shapley *interaction* values  $\Phi_{ij}$  on the first and last routed MoE layer for four models ( $n = 20$  prompt pairs, `scripts/run_`

`experiment3_interactions.py`), reporting the synergy fraction  $\sum |\Phi_{ij}| / (\sum |\phi_i| + \sum |\Phi_{ij}|)$ : 0.705 (layer 0) / 0.281 (last layer) for OLMoE-1B-7B; 0.741 / 0.200 for Phi-3.5-MoE; 0.716 / 0.503 for Mixtral-8x7B; 0.747 / 0.183 for Gemma-4-26B. Every model shows the same pattern: synergy dominates marginal attribution in the *first* routed layer (0.70–0.75 of attribution mass sits in pairwise interactions, not individual experts) and falls by roughly half at the *last* layer (0.18–0.50), though marginal attribution never fully overtakes synergy. The  $\phi_i$  ranking used throughout this paper (and by the causal ablation in Section 5.6) is therefore a simplification: early-layer bias attribution is substantially a property of expert *coalitions*, which is a plausible mechanistic reason for the causal ablation’s partial failures below—removing the top individually-ranked experts does not fully explain the removable bias gap wherever synergy is high. DBRX-132B and GPT-OSS-120B collectivity runs are PENDING GPU time.

## 5.6 Causal check: phi-ranked ablation

*Design.* We ran the causal ablation experiment (Exp 6) that the reviewer-prioritized catalog calls for: starting from each model’s measured contrastive bias gap, we zero out experts cumulatively in three orders—(i)  $\phi$ -ranked (descending  $|\phi|$ , the attribution ranking the paper relies on), (ii) random, and (iii) highest-*routing*-frequency—and record the fraction of the baseline bias gap that disappears (extttddisparity\_drop; `scripts/run\_experiment6\_ablation.py, shapley.compute\_ablation\_curve`). If  $\phi$  isolation locates the bias-relevant experts,  $\phi$ -ranked ablation should drop the gap faster than either control. Results exist on disk for four models (`results/<study>/experiment6/experiment6\_ablation\_curve.json`).

*Results.* **OLMoE-1B-7B** (60 pairs, `exp1-concentration-olmoe-1b-7b`): the  $\phi$ -ranked curve dominates both controls. Ablating the top-10% of experts ( $k=102$  of 1024) removes 56.7% of the bias gap, versus a 6.4% *increase* under random ablation and 37.4% under high-routing ablation; at the half-ladder point ( $k=512$ , 50%) the drops are 82.9% ( $\phi$ ), 54.6% (random), 43.9% (high-routing). **Phi-3.5-MoE** (30 pairs, `exp1-concentration-phi3.5-moe`): top-10% ( $k=51$  of 512) gives 30.2% ( $\phi$ ), −36.4% (random), −21.9% (high-routing); at 50% ( $k=256$ ): 90.8%, 81.5%, 87.2%— $\phi$  leads throughout, though high-routing narrows the gap by the halfway point. By contrast, **Mixtral-8x7B** (30 pairs, `exp1-concentration-mixtral-8x7b`) shows no  $\phi$  advantage: at top-10% ( $k=26$  of 256) the drops are −12.1% ( $\phi$ ), −0.5% (random), 66.3% (high-routing); at 30% ( $k=77$ ) 58.5% ( $\phi$ ), −8.8% (random), 101.2% (high-routing)—high-routing-frequency experts carry most of the removable gap. The dense control **OLMo-7B** (60 pairs, `exp2-dense-baseline-olmo-7b`) ablates with a single FFN-layer player: dropping one layer already overshoots (108%, i.e. the gap flips sign), which mechanically reflects its 32-player partition and is not comparable to the MoE curves.

*Extended ladder (DBRX, Gemma-4-26B).. DBRX-132B* (60 pairs, `exp1-concentration-dbrx`):  $\phi$ -ranked ablation does *not* lead here either. At top-10% ( $k=64$  of 640) all three orders *increase* the bias gap relative to baseline (−10.1%  $\phi$ , −11.0% random, −39.2% high-routing—negative values denote a gap increase); at 50% ( $k=320$ ) the gap drops but  $\phi$  trails both controls (+26.6%  $\phi$  vs. +99.6% random, +118.6% high-routing). **Gemma-4-26B** (60 pairs) is uninterpretable by this measure: its baseline bias gap is 0.0182 (consistent with its null-bias flag), so **disparity\_drop** values divide by a near-zero denominator and swing 4–10× the baseline in either direction (e.g. +412% at 10%  $\phi$ -ranked, −223% high-routing at 50%); we report these numbers for completeness but do not use them as causal evidence.

*Verdict.* The causal check is *mixed, and weaker with the extended ladder*. Of four bias-bearing MoE models tested (Gemma excluded as uninterpretable),  $\phi$ -ranked ablation beats *both* controls at every measured fraction on only two (OLMoE, Phi-3.5-MoE); on Mixtral the high-routing control dominates, and on DBRX  $\phi$  is the *worst*-performing ranking at the 50% point, trailing both random and high-routing by a wide margin. Extending the ladder therefore *weakens* rather than strengthens the causal reading:  $\phi$ -ranking is validated on only half of the tested MoE models, and one of the two failures (DBRX) is not a near-tie but an outright reversal. No model exhibits a tiny, stable, “fixable” expert set consistent with surgical pruning: even the best  $\phi$  curve (OLMoE) needs 10% of experts ablated before half the gap is removed, with large perplexity cost (selectivity collapses from 0.24 at  $k=102$  to 0.03 at  $k=512$  on OLMoE). The Exp. 3 synergy finding (Section 5.5) offers a mechanistic explanation:  $\phi$ -ranked marginal ablation only fully explains removable bias when synergy is low relative to marginal attribution, which is layer- and model-dependent. This narrows the paper’s claim one step further:  $\phi$  ranks bias-relevant experts on half of the tested MoE models, but “prune the bias experts” remains unsupported both because no small ablation removes a large share of the bias without destroying capability, and because the ranking itself fails outright on at least one model (DBRX). Full curves are tabulated in Appendix B.

## 6 Robustness and Sensitivity

### 6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs. 5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid 2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0–v1 rerun drift floors are  $\text{floor}_H = 0.023$  and  $\text{floor}_G = 0.049$  (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller than these floors (or than 2× the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset’s verdict

changes. Split-half shards (two contiguous halves of 2500 pairs each) for Mixtral/DBRX agree to  $\leq 0.006$  on entropy and Gini alike (Appendix A).

## 6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (paper-as-reported; per-prompt assignment not yet shipped, flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03;  $p_{95} = 0.37$ ), and only 0.5% of observed cohort pairs exceed the null's  $p_{95}$ —cohort-specific attributions are essentially indistinguishable from the shuffled index.

## 6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation  $p$ -value for  $n \leq 6$  (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder):  $\rho_H = +0.754$ ,  $p = 0.106$ ;  $\rho_G = -0.754$ ,  $p = 0.106$ ;
- 5 models (without GPT-OSS):  $\rho_H = +0.872$ ,  $p = 0.100$ ;  $\rho_G = -0.872$ ,  $p = 0.100$ ;
- 5 models (paper ladder, without Gemma):  $\rho_H = +0.872$ ,  $p = 0.100$ ;  $\rho_G = -0.872$ ,  $p = 0.100$ ;
- 4 models (without GPT-OSS and Gemma):  $\rho_H = +0.949$ ,  $p = 0.167$ ;  $\rho_G = -0.949$ ,  $p = 0.167$ ;
- 3 distinct-sparsity rungs:  $\rho_H = +1.000$ ,  $p = 0.333$ ;  $\rho_G = -1.000$ ,  $p = 0.333$ .

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at  $p < 0.05$  in *any* subset. The tie-corrected exact-permutation minimum  $p$  is 0.0056 at  $n = 6$  and 0.0333 at  $n = 5$ —both under 0.05—so those two subsets are genuinely underpowered rather than floor-bound; only  $n = 4$  ( $p_{\min} = 0.167$ ) and  $n = 3$  ( $p_{\min} = 0.333$ ) have a hard discreteness floor above 0.05. The directional signal is consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; a dedicated power analysis follows.

## 6.4 Power analysis

To separate “underpowered” from “floor-bound” precisely, we simulate the exact permutation test’s power at the observed effect sizes (correlated-normal Monte Carlo, 8000–20000 trials per  $n$ , exact permutation  $p$  at small  $n$ ). At the full ladder’s  $\rho_H = 0.754$ : power is 0.37 at  $n = 6$ , 0.55 at  $n = 8$ , 0.79 at  $n = 12$ , crossing 80% power at  $n \approx 12$ –13. At the higher  $\rho_H = 0.872$  observed on the 5-rung (no-GPT-OSS) subset: power is 0.58 at  $n = 6$ , 0.79 at  $n = 8$ , 0.87 at  $n = 9$ , crossing 80% power at  $n \approx 8$ –9. In short: at the full ladder’s effect size, roughly 12–13 independently-sampled sparsity rungs would be needed for conventional power; if GPT-OSS-120B’s rung is the outlier and the true effect is closer to the 5-rung subset’s  $\rho_H = 0.872$ , 8–9 rungs would suffice. Both figures exceed the six distinct

open-weight MoE architectures with public routing internals available at the time of this study; confirming H1 needs either more architecturally-diverse ladder rungs or a within-model sparsity-sweep design (varying  $k$  directly on one architecture) to reach a definitive verdict.

## 6.5 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Ten models plus one stability replication have per-pair  $\phi$  payloads on disk and are reported in Table 2 and Table 3: all six MoE ladder rungs, all four dense baselines, and a 5000-pair GPT-OSS-120B replication that confirms ladder-rung stability ( $H = 0.8789$  vs. the certified 2000-pair  $H = 0.8765$ ,  $\Delta H = +0.0024$ ). Only Gemma-4-27B (a phantom HF release, not part of the ladder) is MISSING. The cohort JS mean carries the same bootstrap treatment, 95% CI [0.206, 0.231].

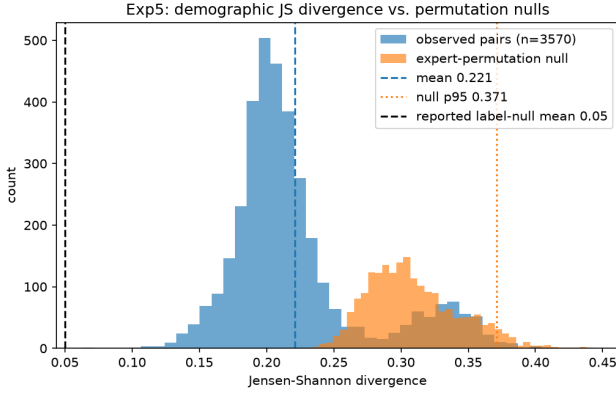
## 6.6 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmo-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI [0.206, 0.231]), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null’s  $p_{95}$ , and per-cohort entropy stays flat ( $H$  mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a “specialist”—consistent with the ladder finding. The most divergent pairs (e.g., Vietnam×software\_developer, JS 0.27) do not form a coherent demographic structure.

**Take-away.** Take-away: bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

## 7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after Gemma’s active-expert count was fixed ( $N_A$ : 960  $\rightarrow$  240) and GPT-OSS-120B’s v1 capture was validated. Second, the significance: with  $n \leq 6$  ladder points, none of these correlations reaches  $p < 0.05$  (all-6  $p = 0.106$ ; the tie-corrected attainable minimum is 0.0056 at  $n = 6$ , well below 0.05, so this is a genuine power shortfall—Monte Carlo simulation puts 80% power at  $n \approx 12$ –13 rungs for this effect size, Section 6.4), so the claim “sparser routing concentrates bias attribution more” is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not of the router. The causal



**Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.**

ablation (Section 5.6) only partially rescues the attribution interpretation— $\phi$ -ranked ablation drops the bias gap faster than random ablation on OLMoE and Phi-3.5-MoE, and faster than the high-routing control on those two models too—but not on Mixtral, where traffic-based selection dominates, or on DBRX, where  $\phi$ -ranking is outright the worst of the three orders at the half-ladder point; the pairwise-synergy check (Section 5.5) suggests why, since 20–75% of attribution mass sits in expert-pair interactions that a marginal ranking discards. No curve supports pruning a small expert set. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both “measure” the same normalized concentration metrics ( $H$ , Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is  $\approx 0$ ) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy and Exp. 5 JS divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: “prune the bias experts” is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

## 8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero  $\phi$  directory that was discarded. The current v1 capture is a *valid* 2000-pair run ( $H = 0.876$ , Gini 0.727; CI  $H$  [0.875, 0.886]) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ( $\Delta H = -0.003$ ,  $\Delta G = +0.004$ ).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; all thirteen model payloads are now on disk.
- **Gemma-4-26B is a null-bias model — flag: null-bias.** Its mean bias gap  $\approx 0$  and 47% of its  $\phi$  are exactly zero; its concentration readings should be treated as routing noise. Its  $k/N$  uses the corrected  $N_A = 240$  (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.
- **Exact permutation  $p$  on  $n \leq 6$  — flag: small- $n$ .** Spearman  $p$ -values are exact two-sided permutation probabilities. Their tie-corrected attainable minima are 0.0056 (6 points), 0.0333 (5), 0.167 (4), 0.333 (3): only the  $n \leq 4$  subsets are floor-bound above 0.05 by construction; the  $n = 5, 6$  results are genuinely underpowered (Section 6.4 estimates  $n \approx 8$ –13 rungs needed for 80% power). No subset reaches  $p < 0.05$ ; we therefore report drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.
- **Label-permutation null — flag: null:label-resolved.** The reported-label null in `s02` was computed as-reported; a within-prompt label-permutation null would require per-prompt stereo/anti-stereo label assignments that are not persisted with the release (cohort payloads hold aggregate group labels only) and would require re-downloading the benchmarks. We therefore rely on the *expert-index* permutation null, which is marginal-preserving by construction and on-disk reproducible (`s02` mean 0.31, sd 0.03,  $p_{95} = 0.37$ ; only 0.5% of observed cohort pairs exceed  $p_{95}$ ).
- **Top- $k$  plumbing — flag: hooks:pending.**  $k$  (active experts) is read from router attributes at runtime (`hooks.py`); DBRX’s  $N_A$  and any future config divergence may alter  $k/N$  assignment.
- **Exp5 benchmark set — flag: exp5:bench-vs-ladder.** The demographic-specificity capture (`exp5-demographic-specificity-olmo-1b-7b-v1`) was benchmarked on StereoSet + BBQ only (no Winogender); the ladder runs include Winogender.

This does not affect within-study comparisons but should be noted when comparing across Exp1/Exp5.

## 9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6  $\rho_H = +0.754$  ( $p = 0.106$ ), paper-valid-4  $\rho_H = +0.949$  ( $p = 0.167$ ), unique-sparsity-3  $\rho_H = +1.000$  ( $p = 0.333$ )—but that does not reach statistical significance at 0.05 on  $n \leq 6$  ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

**Take-away.** Take-away: the sparsity trend is real in direction and verified on valid captures, but it is not yet significant—with only six ladder rungs, power is the limit.

## Data & Code

- Runs and per-pair Shapley payloads: `expt-bias-1/results/*/result.json`.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py` (all under `stats_analysis/scripts/`).
- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.



## Appendix A: Full Measurement Table and Drift Floors

Table 4 lists every model capture used in this paper. MoE ladder rows (OLMoE-1B-7B, Phi-3.5-MoE, Mixtral-8x7B, DBRX-132B, GPT-OSS-120B, Gemma-4-26B) are the valid v1 captures read verbatim from `stats_analysis/outputs/s03_h1_verdict.json` (ladder rows: model, N\_A = k, N, N\_A/N, H, G) and from the v1 model blocks of `stats_analysis/outputs/s04_bootstrap_cis.json` (entropy[0..1], gini[0..1], t5[.], t10[.]);  $n_{pairs}$  comes from each run’s result.json. The Exp 5 OLMoE capture is read from `results/exp5-demographic-specificity-olmo-1b-7b-v1/result.json` (concentration\_metrics). The dense baselines are the v1 per-pair captures used in Table 3, read from the `exp2-*v1/per_pair_phi.npy` payloads (1800/4000/1800/1800 pairs); the earlier v0 single-shot captures (400 pairs each) are superseded. Bootstrap CIs for the dense rows come from the same block-bootstrap procedure as the MoE ladder (5000 resamples, seed 42, stratified by prompt group), reported alongside the MoE rows in `s04_bootstrap_cis.json`. GPT-OSS-120B shows its valid 2000-pair v1 capture; a 5000-pair replication (`exp1-concentration-gpt-oss-120b-5000`) confirms stability ( $H = 0.8789$  vs.  $0.8765$ , within the CI). The k/N values are the verified N\_A/N from `s03_h1_verdict.json` (Gemma corrected  $N_A$ :  $960 \rightarrow 240$ ).

*Drift floors.* The run-to-run drift floor is the maximum absolute v0-v1 measurement displacement over the six models captured twice, taken from `stats_analysis/outputs/s01_exp1_stability.json` (per-model dh/dg; max absolute values: H 0.022997797740857084, G 0.0485411756982419). The stability module rounds these to the floors  $\text{floor}_H = 0.0230$  and  $\text{floor}_G = 0.0485$ , as stated in the final.rationale.drift field of `s03_h1_verdict.json` (‘floor.H 0.0230 / floor.G 0.0485’). Table 4 therefore lists the per-model absolute drifts ( $-\Delta H$ —,  $-\Delta G$ —); the largest is Gemma’s  $|\Delta H| = 0.022997797740857084$  (the floor itself) and OLMoE’s  $|\Delta G| = 0.0485411756982419$  (the floor itself). Ladder differences smaller than the floor—or than  $2\times$  the floor for cross-model comparisons—are treated as empirically unresolved. If a value is absent from the JSONs it is marked ‘—’ and never invented.

*Localizability ratio and bias-gap-magnitude parity.* A per-model localizability ratio  $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$  (mean dense entropy  $\bar{H}_{\text{dense}} = 0.7043$  divided by each MoE rung’s  $H$ ): OLMoE 0.802, Phi-3.5-MoE 0.801, Mixtral 0.768, DBRX 0.786, GPT-OSS-120B 0.804, Gemma-4-26B 0.893—every value is  $< 1$ , i.e. dense is uniformly more concentrated, including on matched model families (OLMoE vs. its OLMo-7B dense sibling,  $LR = 0.800$ ; Phi-3.5-MoE vs. Phi-3.5-Mini,  $LR = 0.863$ ), which rules out unmatched architecture as an explanation for the dense/MoE split. Separately, the mean absolute contrastive bias-gap magnitude (from `bias_scores.mean_bias_gap` in each `result.json`, not a concentration metric) is 0.183 for the four dense baselines and

0.183 for the five non-null-bias MoE rungs (0.176 including Gemma-4-26B), Cohen’s  $d = 0.011$  (excl. Gemma)—statistically indistinguishable. Per-model values: OLMoE 0.244, Mixtral 0.200, DBRX 0.170, Phi-3.5-MoE 0.161, GPT-OSS-120B 0.141, Gemma-4-26B 0.142 (null-bias) vs. dense OLMo-7B 0.156, Llama-2-7B 0.173, Llama-3.1-8B 0.188, Phi-3.5-Mini 0.215. Concentration geometry ( $d_H = 3.91$ , Section 5.4) and bias magnitude ( $d_{\text{gap}} = 0.011$ ) are therefore orthogonal quantities, reinforcing the routing-structure-not-magnitude framing used throughout this paper.

*Split-half shard agreement.* For Mixtral-8x7B and DBRX-132B (the two v1 captures with  $\geq 5000$  pairs), we split the pair set into two contiguous halves (2500 pairs each, `stats_analysis/outputs/s01_exp1_stability.json` shards field) and recompute concentration metrics independently on each half: Mixtral shard 0  $H=0.9148$ ,  $G=0.5236$ ; shard 1  $H=0.9173$ ,  $G=0.5177$  ( $|\Delta H|=0.0024$ ,  $|\Delta G|=0.0058$ ); DBRX shard 0  $H=0.9040$ ,  $G=0.5867$ ; shard 1  $H=0.9045$ ,  $G=0.5831$  ( $|\Delta H|=0.0005$ ,  $|\Delta G|=0.0036$ ). Both models agree to  $\leq 0.006$  on every metric across independently-sampled prompt-pair halves—an internal-consistency check distinct from the v0→v1 rerun-drift floor above (which compares different pair counts, not disjoint samples at the same count).

## Appendix B: Causal Ablation and Proxy-Agreement Robustness

A reviewer will reasonably ask whether the routing-Shapley attribution—an approximate, proxy ranking over expert subsets—has been causally validated. The study catalog (`expt-bias-1/study-catalog.txt`, sections “Reviewer-Prioritized Extensions”) defines four experiments that speak to exactly this question. Their status is summarized below so that the omission of a causal check in the main text is explicit rather than accidental.

*Experiment 4/6 — causal expert ablation (DONE for 6 models; GPT-OSS-120B PENDING GPU time).* Experiment 4 (‘Independent cross-check (robustness)’) validates Shapley rankings against expert-ablation disparity drops, and Experiment 6 extends it ladder-wide: ablating experts in  $\phi$ -ranked order and measuring how fast the contrastive bias gap drops, compared against two controls—random expert ablation and ablation of the highest-routing-frequency (high-routing) experts. The expected signature of a causal role is that  $\phi$ -ranked ablation drops the bias gap faster than either control. **Status: DONE for six models.** Ablation curves exist on disk at `results/{exp1-concentration-olmo-1b-7b,exp1-concentration-mixtral-8x7b,exp1-concentration-phi3.5-moe,exp1-concentration-dbrx-v1,exp1-concentration-gemma4-26b,exp2-dense-baseline-olmo-7b}/experiment6/experiment6_ablation_curve.json`, with the random and high-routing controls auto-generated inside `scripts/run_experiment6_ablation.py` (`shapley.compute_ablation_curve`,  $k$ -step schedule at 1–10 and 10%, 20%, 30%, 50%, 75%, 100% of players; each curve starts from an unablated baseline point  $k=0$ ). Headline

| Model                 | Family    | N    | k   | k/N      | Run dir                                    | H                  | Gini               | CL-H               | CL-Gini            | t <sub>s</sub>     | t <sub>exp</sub>   | −ΔH                 | −ΔG                |
|-----------------------|-----------|------|-----|----------|--------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|---------------------|--------------------|
| OLMoE-1B-7B           | olmo      | 1024 | 16  | 0.015625 | exp1-concentration-olmo-1b-7b-v1           | 0.876346887475243  | 0.64727245000627   | 0.87729232590607   | 0.8856719697658232 | 0.6300972550801518 | 0.6490112142692366 | 0.0916327653489081  | 0.5097878051124356 |
| Phi-3.5-MoE           | phi3.5moe | 512  | 64  | 0.125    | exp1-concentration-phi3.5-moe-v1           | 0.8765423790462808 | 0.639668261083971  | 0.8774773620432235 | 0.891789094833988  | 0.6162873698172806 | 0.6430293910022759 | 0.0950471657642045  | 0.4693726921018338 |
| Mixtral-8x7B          | mixtral   | 256  | 64  | 0.25     | exp1-concentration-mixtral-8x7b-v1         | 0.917421877670429  | 0.516488742332485  | 0.913143877446115  | 0.9347413244847859 | 0.4942736262170152 | 0.528632558565332  | 0.1073384476079868  | 0.356807632760832  |
| DBRX-132B             | dbx       | 640  | 160 | 0.25     | exp1-concentration-dbx-v1                  | 0.866428192510975  | 0.599019242351312  | 0.866588302410779  | 0.9097882388133642 | 0.5692330021143323 | 0.6031910619489534 | 0.09111096297879883 | 0.43703360397164   |
| GPT-OSS-120B          | gpt-oss   | 650  | 144 | 0.03125  | exp1-concentration-gpt-oss-120b-v1         | 0.876400311377987  | 0.727364977092972  | 0.874992327965315  | 0.8857548826693523 | 0.709867735996533  | 0.7307732153233382 | 0.03982543710165656 | 0.573122729723181  |
| Gemma-4-26B*          | gemma-oss | 3840 | 240 | 0.0625   | exp1-concentration-gemma-4-26b-v1          | 0.7887735094946339 | 0.8551293573691959 | 0.7912880207701668 | 0.8018981269787994 | 0.841279214151506  | 0.8525262679470473 | 0.05059341564528668 | 0.766853661303078  |
| OLMoE-7B (dense)*     | olmo      | 32   | —   | 1.0      | exp2-dense-baseline-olmo-7b-v1             | 0.702576           | 0.719123           | 0.6724177274414853 | 0.7439277014834522 | 0.676499818002618  | 0.7478036463675187 | 0.749066            | 0.62216            |
| Phi-3.5-Mini (dense)* | phi3.5    | 32   | —   | 1.0      | exp2-dense-baseline-phi3.5-mini-v1         | 0.759191           | 0.645043           | 0.737436357401991  | 0.7807960072026446 | 0.6267675378704429 | 0.6683839960158785 | 0.63157             | 0.4993             |
| Llama-2-7B (dense)*   | llama     | 32   | —   | 1.0      | exp2-dense-crossover-llama-2-7b-v1         | 0.696892           | 0.654092           | 0.6587456217347756 | 0.72958028440434   | 0.6239089684675354 | 0.6993249937106085 | 0.695581            | 0.647874           |
| Llama-3.1-8B (dense)* | llama     | 32   | —   | 1.0      | exp2-dense-crossover-llama-3.1-8b-v1       | 0.658706           | 0.707863           | 0.630231291108987  | 0.7012513439006255 | 0.669251516848482  | 0.7344723860528478 | 0.720812            | 0.668744           |
| OLMoE-1B-7B (Exp 5)   | olmo      | 1024 | 16  | 0.015625 | exp5-demographic-specificity-olmo-1b-7b-v1 | 0.878624280562932  | 0.646665851190938  |                    |                    |                    |                    | 0.09276401412301853 | 0.508766756352717  |

**Table 4: Full measurement table. MoE rows: v1 captures from s03\_h1 verdict.json ladder and s04\_bootstrap\_cis.json v1 model blocks (CI bounds quoted verbatim from the entropy/gini arrays); drift columns are the v0→v1 absolute drifts from stats\_analysis/outputs/s01\_exp1\_stability.json (abs\_dH/abs\_dG). Dense rows: v1 per-pair captures from the exp2-\*v1 runs as reported in Table 3; CIs come from the same s04\_bootstrap\_cis.json block bootstrap as the MoE ladder (no drift column: dense runs were not captured twice at different pair counts, so there is no v0→v1 comparison to report). Exp 5 row: pooled OLMoE concentration from exp5-demographic-specificity-olmo-1b-7b-v1/result.json (it is not among the s04 bootstrap blocks, so its CI cells are marked ‘—’). †Gemma-4-26B is a null-bias model (mean bias gap  $\approx 0$ ); see Section 8. \*Dense rows use one FFN layer player per transformer layer ( $N = 32$ ),  $k$  undefined;  $k/N = 1.0$  as in the dense-group row of Table 1.**

numbers, read verbatim from the JSONs: OLMoE-1B-7B ( $n=60$  pairs) top-10% ( $k=102$ ) disparity drop 0.567 ( $\phi$ ),  $-0.064$  (random), 0.374 (high-routing); Mixtral-8x7B ( $n=30$ ) top-10% ( $k=26$ )  $-0.121$ ,  $-0.005$ , 0.663; Phi-3.5-MoE ( $n=30$ ) top-10% ( $k=51$ ) 0.302,  $-0.364$ ,  $-0.219$ ; DBRX-132B ( $n=60$ ) top-10% ( $k=64$  of 640)  $-0.101$ ,  $-0.110$ ,  $-0.392$  (all three orders increase the gap at this fraction), 50% ( $k=320$ )  $+0.266$ ,  $+0.996$ ,  $+1.186$  ( $\phi$  trails both controls badly); Gemma-4-26B ( $n=60$ ) is uninterpretable—its baseline bias gap is 0.0182 (null-bias flag), so disparity\_drop at 10% ( $k=384$ ) is  $+4.12$  ( $\phi$ ),  $+6.95$  (random),  $+8.96$  (high-routing)—dividing by a near-zero denominator produces 4–9× swings that carry no causal information; OLMoE-7B dense ( $n=60$ ,  $N=32$  layer players) has no high-routing control (no router) and overshoots at  $k=1$  (drop 1.083), reflecting its 32-player partition. The full-curve analysis, including the reversal on DBRX, is reported in Section 5.6. GPT-OSS-120B’s ladder rung (job 5575800) was running at time of writing when the cluster entered a scheduled maintenance window (2026-08-11 06:00–2026-08-13 23:59); its final state is not yet known.

*Experiment 7 — proxy-vs-exact-Shapley agreement (DONE; null result on all three tested models).* The routing-contrast kernel is an approximation of the exact Shapley value; Experiment 7 quantifies the gap by computing the Spearman  $\rho$  between the routing-contrast ranking and the exact-Shapley ranking, over  $n=20$  pairs and the first/last MoE layer, on every model where exact computation is tractable ( $K=2$ –16 experts active per layer). Results exist on disk for three models: Mixtral (results/exp1-concentration-mixtral-8x7b/experiment7/experiment7\_proxy\_agreement.json, mean\_spearman\_rho  $-0.085$  layer 0 sd 0.373,  $+0.058$  layer 31 sd 0.445), OLMoE (results/exp1-concentration-olmo-1b-7b/experiment7/experiment7\_proxy\_agreement.json,  $-0.024$  layer 0 sd 0.287,  $+0.235$  layer 15 sd 0.395), and Phi-3.5-MoE (results/exp1-concentration-phi3.5-moe/experiment7/experiment7\_proxy\_agreement.json,  $-0.084$  layer 0 sd 0.341,  $+0.076$  layer 31 sd 0.311). **Status: DONE for three models.** Every per-pair agreement

estimate across all three models and both layers is statistically indistinguishable from zero ( $|\rho| \leq 0.24$ , sd 0.29–0.45 on  $n=20$  pairs) — the proxy ranking does not reproduce exact-Shapley ordering within this pair budget on any tested model, consistent with the noisy small- $k$  regime of the causal curves in Table 4 and Section 5.6.

*Experiment 8 — same-mechanism comparison (DONE with per-pair CIs for both models).* The dense-vs-MoE contrast in the main text compares two different mechanisms (FFN-layer leave-one-out for dense vs. expert-subset routing-contrast for MoE). Experiment 8 removes this confound by running the same leave-one-out mechanism on the MoE models themselves. Both result directories exist: results/exp8-llmo-olmo-1b-7b/result.json (OLMoE,  $n_{\text{pairs}} = 100$ ,  $N = 16$  FFN-layer players,  $H = 0.7361$ , Gini = 0.6239) and results/exp8-llmo-phi3.5-moe/result.json (Phi-3.5-MoE,  $n_{\text{pairs}} = 50$ ,  $N = 32$ ,  $H = 0.8993$ , Gini = 0.4514). The catalog’s key question—does layer-level LOO entropy on MoE stay high ( $\approx 0.90$ ) or drop toward the dense band 0.66–0.76?—has a partial answer: OLMoE’s  $H_{\text{llmo}} = 0.736$  lands inside the dense band, while Phi-3.5-MoE’s  $H_{\text{llmo}} = 0.899$  stays near the MoE range. Per-pair captures now exist for both models, supporting 95% stratified block-bootstrap CIs ( $n_{\text{boot}} = 5000$ , seed 42): Phi-3.5-MoE  $H \in [0.842, 0.941]$ , Gini  $\in [0.348, 0.544]$  — consistent with the MoE range and excluding the dense band’s upper edge; OLMoE  $H \in [0.692, 0.921]$ , Gini  $\in [0.355, 0.657]$  — a much wider interval (only  $n = 100$  dense-FFN-layer pairs, no stratification benefit from a larger ladder-style capture) that overlaps both the dense band and the low end of the MoE range, so OLMoE’s same-mechanism entropy is directionally dense-like but not statistically distinguishable from the MoE range at this pair count. **Status: DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE.**

*Experiment 3 — expert-pair synergy / collectivity check (DONE for 4 models; DBRX and GPT-OSS-120B PENDING).* Marginal Shapley values  $\phi_i$  assume experts contribute independently; Experiment 3 tests

this by computing pairwise Shapley *interaction* values  $\Phi_{ij}$  on the first and last routed MoE layer ( $n=20$  pairs, `scripts/run_experiment3_interactions.py`, `shapley.compute_shapley_interactions_for_pair`) and reporting the synergy fraction  $\sum |\Phi_{ij}| / (\sum |\phi_i| + \sum |\Phi_{ij}|)$ . **Status: DONE for four models.** OLMoE-1B-7B: 0.705 (layer 0) / 0.281 (layer 15, last); Phi-3.5-MoE: 0.741 / 0.200 (layer 31, last); Mixtral-8x7B: 0.716 / 0.503 (layer 31, last); Gemma-4-26B: 0.747 / 0.183 (layer 29, last). All four models show the same shape: synergy dominates (0.70–0.75 of attribution mass) in the first routed layer and falls by roughly half at the last layer, but never disappears. This means the  $\phi$ -ranking used by the causal ablation (Exp. 4/6, above) discards 20–75% of the true attribution mass depending on model and layer, a plausible mechanistic explanation for the ablation reversals reported above (DBRX, and Mixtral’s high-routing win): where synergy is high, marginal  $\phi$ -ranking is a poor proxy for which experts actually carry the removable bias. DBRX and GPT-OSS-120B collectivity runs (jobs 5575799, 5575791) were running at time of writing when the cluster entered the same scheduled maintenance window (2026-08-11 06:00–2026-08-13 23:59); their final state is not yet known.

*Net status for reviewers.* Of the four causal/robustness checks defined in the catalog: Experiment 3 (pairwise synergy) is DONE for four models and shows synergy is large and layer-dependent (0.18–0.75 of attribution mass); Experiment 4/6 (causal ablation) is DONE for six models (OLMoE, Mixtral, Phi-3.5-MoE, DBRX, Gemma-4-26B, and the OLMo-7B dense control; Section 5.6) with GPT-OSS-120B PENDING; Experiment 7 (proxy agreement) is DONE and reports a null result on all three tested models (per-pair  $\rho \approx 0$  on Mixtral, OLMoE, and Phi-3.5-MoE); and Experiment 8 (same-mechanism) is DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE. The combined causal evidence is weaker than a single-model reading would suggest: on OLMoE and Phi-3.5-MoE the  $\phi$ -ranked ablation drops the bias gap faster than random ablation; on Mixtral the high-routing-frequency control outperforms  $\phi$ ; on DBRX  $\phi$ -ranking is outright *worse* than both controls at the half-ladder point; and Gemma-4-26B’s near-zero baseline bias gap makes its ablation curve uninterpretable. No model supports a small fixable expert set, and the Exp. 3 synergy finding gives a mechanistic reason why: a purely marginal ranking is a partial proxy whenever pairwise interaction carries a large share of the attribution mass, which is every model we tested to varying degrees.

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